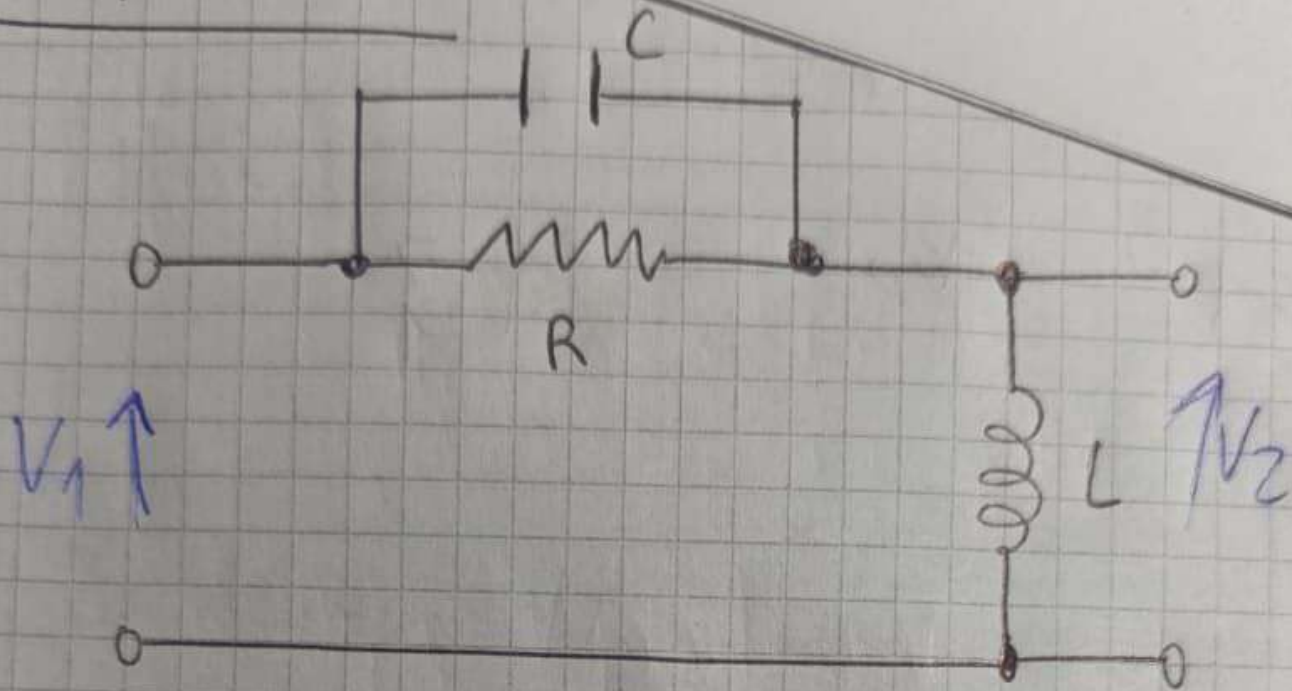
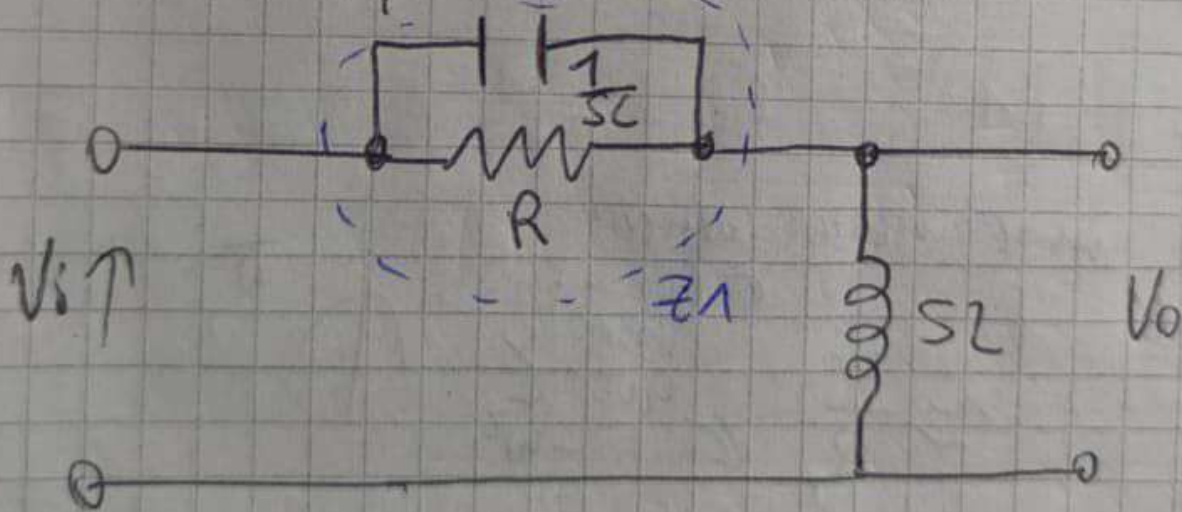


CIRCUITO B



Circuito equivalente:



$$Z_1 = R \parallel \frac{1}{sC} = \frac{R}{sCR + 1}$$

$$\frac{1}{Z_1} = \frac{1}{R} + sC = \frac{1 + sCR}{R}$$

$$Z_T = Z_1 + SL = \frac{R}{sCR+1} + SL = \frac{R + s^2 LCR + SL}{sCR+1}$$

Aplico un divisor de Tensión para Hallar la función de transferencia

$$V_o(s) = V_i(s) \cdot \frac{SL}{Z_T}$$

$$H(s) = \frac{V_o(s)}{V_i(s)} = SL \cdot \frac{sCR+1}{s^2 LCR + SL + R} = \frac{s^2 LCR + SL}{s^2 LCR + SL + R}$$

$$H(s) = \frac{SL(sRC+1)}{s^2 LCR + SL + R}$$

Divido todo por RLC

$$H(s) = \frac{s^2 + s \frac{1}{RC}}{s^2 + s \frac{1}{RC} + \frac{1}{LC}} = \frac{s^2 + s \frac{\omega_0}{Q}}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}$$

$$\omega_0^2 = \frac{1}{LC} \Rightarrow LC = 1 = \frac{1}{\omega_0^2} \Rightarrow L = \frac{1}{C} = \frac{1}{Q}$$

$$\frac{\omega_0}{Q} = \frac{1}{RC} \Rightarrow RC = \frac{Q}{\omega_0} \Rightarrow C = \frac{Q}{R \cdot \omega_0} \Rightarrow \boxed{C=Q} \quad C=Q$$

$$\left[Q = \frac{\sqrt{2}}{2} \Rightarrow C = \frac{\sqrt{2}}{2} \text{ F} \quad L = \sqrt{2} \text{ H} \right]$$

Entonces

$$H(s) = \frac{s^2 + \sqrt{2}s}{s^2 + \sqrt{2}s + 1} = \frac{s(s + \sqrt{2})}{s^2 + \sqrt{2}s + 1}$$

$$H(j\omega) = \frac{(j\omega)^2 + \sqrt{2}j\omega}{(j\omega)^2 + \sqrt{2}j\omega + 1}$$

$$j\omega^2 = -\omega^2$$

$$H(j\omega) = \frac{-\omega^2 + j\sqrt{2}\omega}{-\omega^2 + j\sqrt{2}\omega + 1} = \frac{-\omega^2 + j\sqrt{2}\omega}{1 - \omega^2 + j\sqrt{2}\omega}$$

$$|H(j\omega)| = \frac{|-\omega^2 + j\sqrt{2}\omega|}{|1 - \omega^2 + j\sqrt{2}\omega|}$$

Respuesta de Módulo

$$|H(j\omega)| = \frac{\sqrt{(-\omega^2)^2 + (\sqrt{2}\omega)^2}}{\sqrt{(1 - \omega^2)^2 + (\sqrt{2}\omega)^2}} = \frac{\sqrt{\omega^4 + 2\omega^2}}{\sqrt{(1 - \omega^2)^2 + 2\omega^2}} =$$

$$|H(j\omega)| = \sqrt{\frac{\omega^4 + 2\omega^2}{(1 - \omega^2)^2 + 2\omega^2}} = \sqrt{\frac{\omega^4 + 2\omega^2}{1 - 2\omega^2 + \omega^4 + 2\omega^2}} = \sqrt{\frac{\omega^4 + 2\omega^2}{1 + \omega^4}}$$

Cuando $\omega \rightarrow 0$

$$|H(j\omega)| = \frac{0}{1} = 0 \rightarrow \text{En dB tiende a } -\infty$$

Cuando $\omega = 1$

$$|H(j)| = \sqrt{\frac{1+2}{1+1}} = \sqrt{\frac{3}{2}} = 1,225$$

$$20 \log_{10}(1,225) = 1,76 \text{ dB}$$

Cuando $\omega \rightarrow +\infty$

$$|H(j\omega)| \approx \sqrt{\frac{\omega^4}{\omega^4}}$$

$$\lim_{\omega \rightarrow +\infty} |H(j\omega)| = \sqrt{\frac{\omega^4}{\omega^4}} = 1 \rightarrow \text{En dB tiende a } 0.$$

Respuesta de Fase:

$$\angle H(j\omega) = \tan^{-1}\left(\frac{\text{Im}(\text{num})}{\text{Re}(\text{num})}\right) - \tan^{-1}\left(\frac{\text{Im}(\text{denom})}{\text{Re}(\text{denom})}\right)$$

$$\angle H(j\omega) = \tan^{-1}\left(\frac{\omega\sqrt{2}}{-\omega^2}\right) - \tan^{-1}\left(\frac{\omega\sqrt{2}}{1-\omega^2}\right)$$